

Probability Distributions

1. Probability Basics

Probability measures the chance of an event occurring.

Formula:

$$P(\text{Event}) = \text{Favorable Outcomes} / \text{Total Outcomes}$$

Properties:

- $0 \leq P(E) \leq 1$
- $P(\text{Sure Event}) = 1$
- $P(\text{Impossible Event}) = 0$
- $P(\text{Not } E) = 1 - P(E)$

Example:

For a fair die, $P(\text{rolling } 4) = 1/6$.

2. Probability Distribution

A probability distribution describes how probabilities are assigned to values of a random variable.

Types:

- Discrete Distribution
- Continuous Distribution

Functions:

- PMF – Probability Mass Function
- PDF – Probability Density Function
- CDF – Cumulative Distribution Function

3. Bernoulli Distribution

Models a single experiment with two outcomes: Success or Failure.

Parameter:

p = Probability of success

Mean = p

Variance = $p(1-p)$

Examples:

- Coin toss
- Pass/Fail result

4. Binomial Distribution

Represents repeated Bernoulli trials.

Conditions:

- Fixed number of trials
- Independent trials
- Two outcomes
- Constant probability

Formula:

$$P(X=x) = nCx \cdot p^x \cdot (1-p)^{(n-x)}$$

$$\text{Mean} = np$$

$$\text{Variance} = np(1-p)$$

5. Poisson Distribution

Used to count events occurring in a fixed interval.

Formula:

$$P(X=x) = e^{(-\lambda)} \lambda^x / x!$$

$$\text{Mean} = \lambda$$

$$\text{Variance} = \lambda$$

Examples:

- Calls per hour
- Accidents per day

6. Uniform Distribution

Every outcome has equal probability.

Example:

Rolling a fair die.

$$\text{Mean} = (a+b)/2$$

Applications:

- Simulations
- Random number generation

7. Normal (Gaussian) Distribution

A continuous bell-shaped distribution.

Characteristics:

- Symmetric about mean
- Mean = Median = Mode

Empirical Rule:

- $\mu \pm 1\sigma \rightarrow 68\%$
- $\mu \pm 2\sigma \rightarrow 95\%$
- $\mu \pm 3\sigma \rightarrow 99.7\%$

Applications:

- Heights
- Weights
- Exam scores

8. Exponential Distribution

Models waiting time between independent events.

Parameter:

λ = Rate

PDF:

$$f(x) = \lambda e^{(-\lambda x)}$$

Applications:

- Waiting time for buses
- Reliability analysis

9. Central Limit Theorem

As sample size increases, the sampling distribution of the sample mean approaches a normal distribution.

Importance:

- Basis of statistical inference
- Enables normal approximation

10. Covariance

Covariance measures how two variables change together.

Interpretation

↑ ↑ : Positive (+ve)

↓ ↓ : Positive (+ve)

↑ ↓ : Negative (-ve)

↓ ↑ : Negative (-ve)

Meaning

Positive Covariance (>0): Both variables tend to move in the same direction.

Negative Covariance (<0): Variables tend to move in opposite directions.

Covariance = 0: No linear relationship (No correlation). .

11. Correlation

Correlation is the scaled (normalized) version of covariance.

Range: $-1 \leq r \leq 1$

Interpretation

$r = +1$: Perfect positive correlation

$r = 0$: No correlation

$r = -1$: Perfect negative correlation

Formula: $r = \text{Cov}(X,Y) / (\sigma_X \sigma_Y)$

Correlation = Covariance ÷ (Product of Standard Deviations)

Correlation is called scaled covariance because it is standardized between -1 and $+1$.

12. Quick Revision Sheet

Bernoulli → Single trial

Binomial → Multiple Bernoulli trials

Poisson → Count of events

Uniform → Equal probability

Normal → Bell curve

Exponential → Waiting time

CDF → Cumulative probability

68–95–99.7 Rule:

$1\sigma = 68\%$

$2\sigma = 95\%$

$3\sigma = 99.7\%$